Fully neural MS2LDA versus Tomotopy at matched topic capacity

MS2LDA research checkpoint

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Abstract

We evaluate a collapse-resistant, fully neural MS2LDA architecture against the established Tomotopy implementation under a matched protocol: 1,000 requested topics, seed 42, identical spectra, split, vocabulary and preprocessing, and a six-thread CPU allowance. The neural model discovers topics without a Tomotopy teacher, pretrained spectrum encoder, variational Bayes, chemistry labels, or test information. Its checkpoint is fixed in advance to the final training epoch, eliminating validation-gate and checkpoint-selection effects. The Tomotopy model is the previously completed six-worker reference trained with the published MS2LDA parameters; its model and data hashes are verified, and its held-out inference and downstream chemistry are recomputed with six workers. This report presents the resulting likelihood, topic quality, inventory, chemistry, runtime, and memory evidence as a single-seed research comparison rather than a production-replacement claim. Neural held-out NLL is 8.3660 versus 9.7569 for Tomotopy, and mean NPMI is marginally better (-0.2965 versus -0.2996), while cached inference is 259 times faster. Tomotopy retains higher topic diversity, broader active and non-redundant inventory, and substantially higher chemical annotation coverage.

1 Question and comparison contract

MS2LDA transfers the document–topic–word abstraction to tandem mass spectrometry: spectra are documents, fragments and neutral losses are words, and recurring word distributions are Mass2Motifs [1]. The classical model performs iterative inference for each held-out spectrum. The neural model instead learns a direct routing function and produces a topic mixture in one forward pass.

The scientific question is deliberately narrow: when both methods request the same number of topics and receive the same CPU-thread allowance, what quality, chemical coverage, speed, and resource trade-offs remain? The controlled quantities are:

- seed 42 and 1,000 requested topics for both methods;
- identical train, validation, and test matrices and vocabulary;
- six software threads for neural training and inference, Tomotopy training and inference, SGNS, and numerical chemistry stages;
- the published Tomotopy priors $\alpha = 0.6$ and $\eta = 0.1$, parallel scheme 3, and the published convergence procedure; and
- 100 Tomotopy inference iterations, retained as the established held-out evaluation setting.

Equal resources do not mean equal operations. Neural inference is one learned pass; Tomotopy inference performs 100 sampling iterations. The throughput result therefore compares the practical inference procedures supplied by the two methods, not the cost of one mathematically identical update.

2 Data and leakage control

The input is the positive-ion MSnLib benchmark from Zenodo record 20179680. Of 41,568 input spectrum blocks, 38,888 pass parsing, metadata, molecular-structure, and peak filters. Peaks are restricted to $0 < m/z \leq 2000$, relative intensity 0.01–1.0, at most 1,000 fragments, and at least five retained fragments. Fragment and neutral-loss masses are rounded to two decimal places. The training-only vocabulary keeps features occurring in at least three training documents and contains 21,233 words.

Molecules are split by connectivity and Bemis–Murcko scaffold groups so that related spectra remain in one partition. The deterministic 70/10/20 split has 27,222 training, 3,889 validation, and 7,777 test spectra, with zero compound or scaffold-group leakage. Held-out document completion deterministically divides physical peak groups between observed and completion halves. Both methods receive the same matrices in the same column and row order.

3 Neural model

Each vocabulary word combines a 48-dimensional skip-gram embedding learned only from the training corpus, Fourier mass features, and a fragment/loss indicator. A learned projection maps these inputs into the topic geometry. The hierarchical top-2 router combines local token context with a shared document-level score and directly returns the spectrum topic mixture θ. The same learned topic prototypes define the topic–word distribution β.

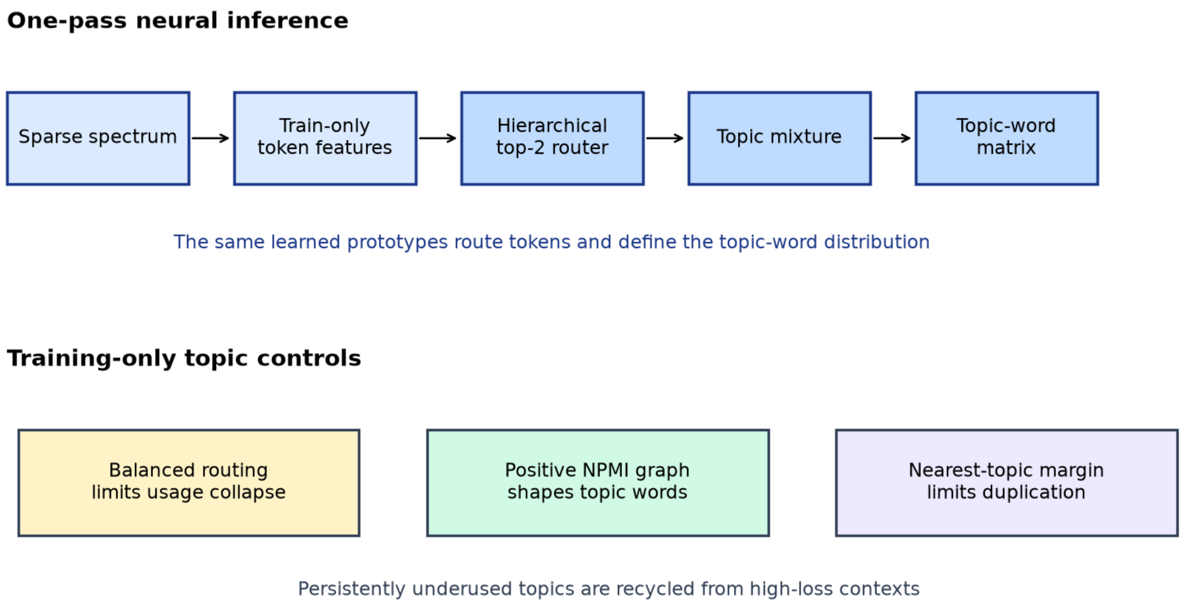

Figure 1: Neural inference and the three complementary training controls. Balanced routing limits usage collapse, the positive-NPMI graph shapes topic words, and a nearest-topic margin limits semantic duplication.

Training uses four deterministic paired views of each training spectrum. The router objective combines held-out-view completion, balanced Sinkhorn targets, and cross-view consistency. Topic updates add a local decoder term, a train-only positive-NPMI co-occurrence constraint, and a nearest-topic margin. Persistently underused topics can be recycled from high-loss contexts.

The model trains for exactly 40 epochs. Validation is measured every two epochs for diagnostics, but it is not compared with thresholds and cannot select a checkpoint. The final epoch-40 checkpoint is written and hashed before the test split is accessed. This fixed-final-epoch rule is the only selection rule.

4 Tomotopy reference and fresh inference

The comparator is the existing K=1000 Tomotopy model trained on the same seed-42 corpus with six workers. It converged after 750 iterations. Reusing it avoids an unnecessary repeat of an already completed training computation. The runner refuses the reference unless its protocol, topic count, seed, training worker count, training parallel scheme, model hash, vocabulary hash, split identifiers, and train/test matrix hashes agree with the current protocol.

Reuse is limited to the trained model. Held-out observed and full-spectrum topic mixtures are recomputed with six workers and 100 inference iterations; completion likelihood, active and effective topic counts, latency, and all theta-dependent chemical results are then regenerated. The source run remains read-only. Tomotopy training time and peak memory are carried forward with their source hashes and are labelled as reused evidence.

5 Evaluation

Predictive fit is the completion negative log-likelihood (NLL) per in-vocabulary token. Topic repetition is summarized by top-10 word diversity. Coherence is normalized pointwise mutual information (NPMI) of top-word pairs measured only on training documents; pairs never observed together score -1, and their fraction is reported separately. Usage is described by the number of corpus-active topics, the median effective topic count per spectrum, and the number of non-overlapping top-10-word equivalents needed to cover 99 percent of corpus topic mass.

Chemical evaluation uses leakage-filtered Mass2Motif Annotation Guidance (MAG). Library compounds sharing held-out connectivity keys are excluded before retrieval. Annotation coverage is the fraction of requested topics with an optimized motif. Structure overlap score (SOS) is averaged first over unique held-out compounds within a topic and then over eligible topics. We report dominant-topic and probability-at-least-0.5 associations.

Cached inference throughput uses 128 spectra, five repeats, and the median seconds per spectrum after one warm-up. Both methods use six software threads. Peak resident memory and wall-clock training time are reported from the corresponding six-worker training runs on the same machine.

6 Results

Table 1: Matched K=1000, six-thread seed-42 comparison.

Metric	Neural	Tomotopy
Topics requested	1000	1000
Held-out NLL (lower is better)	8.3660	9.7569
Top-word diversity	0.7100	0.8145
Mean NPMI	-0.2965	-0.2996
Undefined top-word pairs	43.3%	45.6%
Corpus-active topics	287	363
Median effective topics per spectrum	13.31	9.12
Mass-99 distinct-topic equivalents	663.6	779.9
Annotation coverage	38.4% (384/1000)	60.7% (607/1000)
Dominant SOS-evaluable topics	332/1000	598/1000
Dominant mean SOS	0.6125	0.6238
High-confidence associations	138	961
Six-thread cached spectra/s	7,793.1	30.0
Training time (minutes)	84.4	76.0
Peak resident memory (GiB)	4.07	1.41

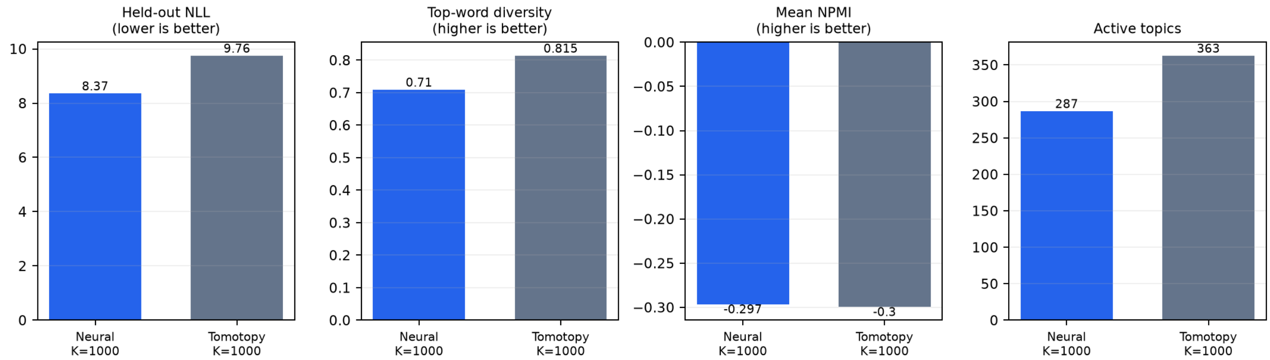

Figure 2: Held-out fit, word diversity, training-corpus NPMI, and active-topic inventory at matched requested capacity. Each panel retains its natural scale; directionality is stated in the title.

At matched topic capacity, the neural model materially improves held-out completion likelihood and reaches slightly better training-corpus coherence: NLL is 8.3660 rather than 9.7569, NPMI is −0.2965 rather than −0.2996, and the undefined-pair fraction is 43.35 percent rather than 45.62 percent. The coherence difference is small, but it removes the large gap observed during early architecture development. Neural cached throughput is 7,793.1 spectra/s versus 30.0 spectra/s for Tomotopy, a 259-fold difference under the same six-thread allowance.

The improvement does not extend to every topic-quality dimension. Tomotopy has higher top-word diversity (0.8145 versus 0.7100), more corpus-active topics (363 versus 287), and a larger mass-99 non-redundant inventory (779.9 versus 663.6 topic equivalents). MAG annotates 607 Tomotopy topics and 384 neural topics. Dominant-topic mean SOS is close (0.6238 versus 0.6125), but Tomotopy also produces many more high-confidence topic–spectrum associations (961 versus 138). Neural training takes 84.4 minutes and peaks at 4.07 GiB resident memory, compared with 76.0 minutes and 1.41 GiB for the Tomotopy source run. The result is therefore a major

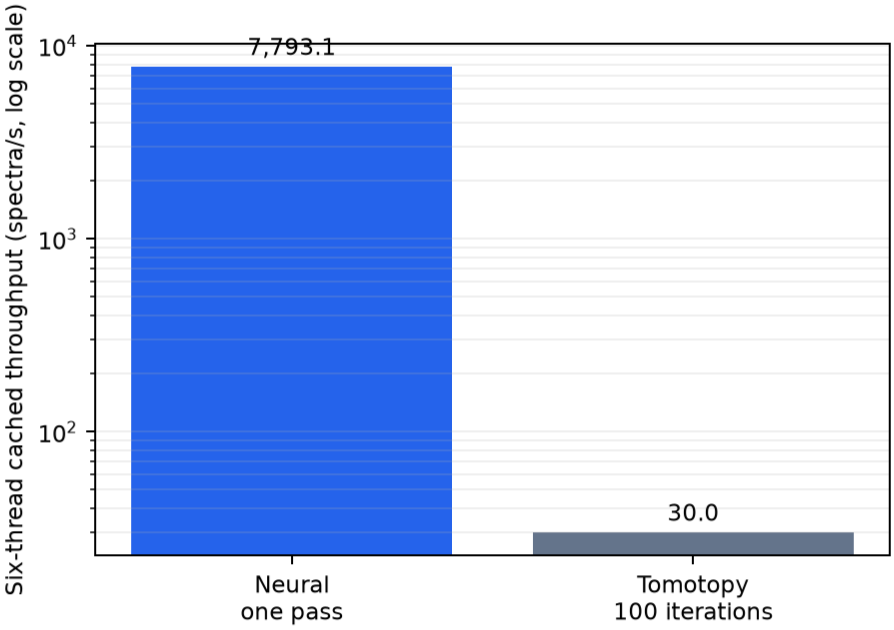

Figure 3: Cached six-thread held-out inference throughput. The logarithmic axis is necessary because the methods perform different native inference procedures.

architecture improvement in likelihood, coherence, and inference speed, but not yet a complete replacement on topic inventory or chemical coverage.

7 Interpretation and limitations

This is an applied single-seed checkpoint, not a general neural-topic-model benchmark. Matching requested capacity and CPU threads removes two important confounders, but the methods retain different optimizers, stopping rules, and inference algorithms. Tomotopy training is reused rather than repeated; its fresh inference and downstream theta-dependent evaluation are nevertheless part of the current run. Cached throughput excludes data loading and reports a batch benchmark rather than end-to-end application latency.

The neural model remains research-only and does not replace the production Tomotopy backend. The present experiment also does not constrain selected topics during inference. A future extension could accept fixed motif distributions alongside free neural topics, mask gradients for the fixed rows, and infer a joint mixture over both groups; this is tracked separately in issue 14 (https://github.com/joewandy/MS2LDA/issues/14).

8 Reproducibility

The workflow records the resolved protocol, source-code hashes, environment, raw-input hashes, comparator-reference evidence, checkpoint hash, test-access record, and output hashes. It resumes neural training exactly. The comparator reference is explicit rather than embedded as a machine-specific default:

```
python -m benchmarks.neural_assignment_ms2lda run \
  --data-root /path/to/MSnLib-assets \
  --run /path/to/run \
  --tomotopy-reference-run /path/to/frozen-tomotopy-run

python -m benchmarks.neural_assignment_ms2lda verify \
  --data-root /path/to/MSnLib-assets \
  --run /path/to/run
```

The committed evidence contains one comparison summary and one portable neural model bundle. Older capacity screens are retained in Git history rather than mixed into the active report.

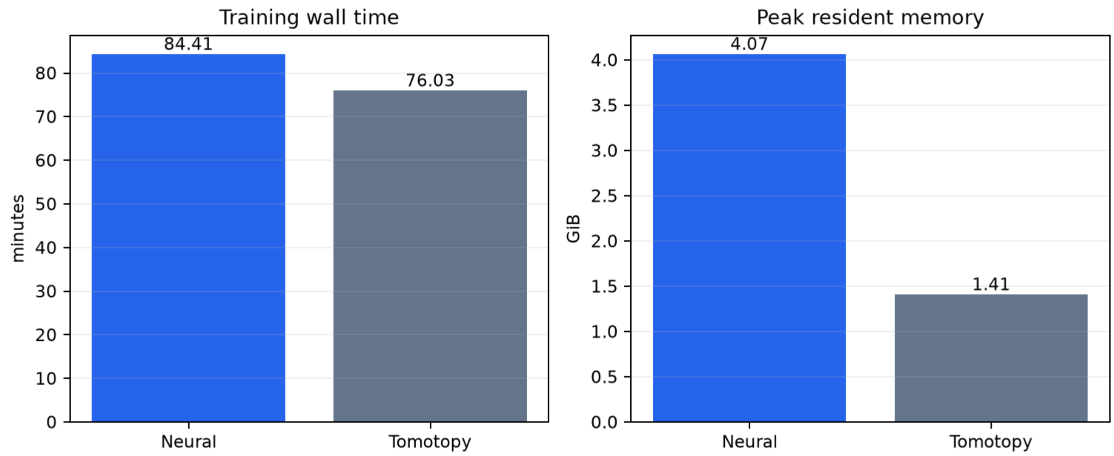

Figure 4: Six-worker training time and peak resident memory. Tomotopy values come from the hash-verified source training run; neural values come from the fresh run.

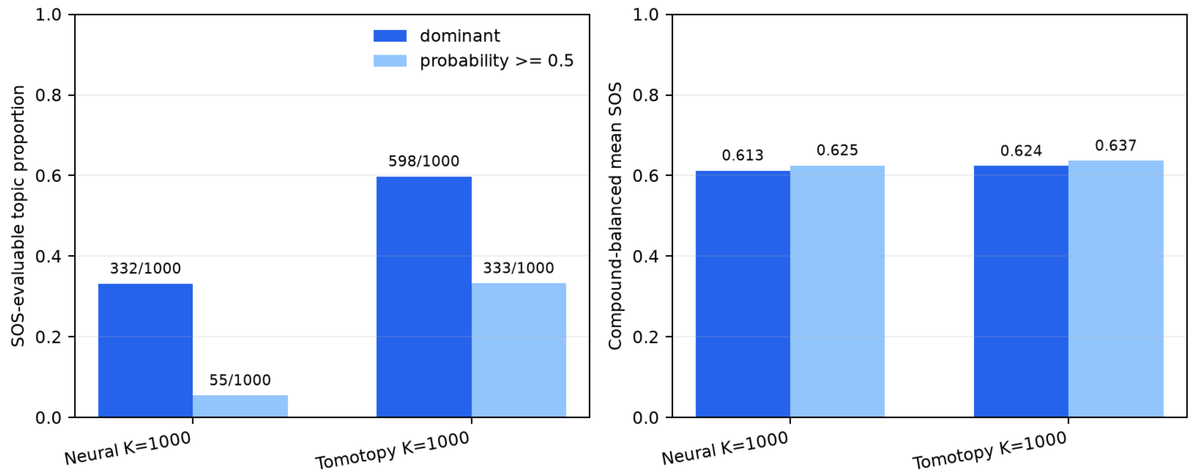

Figure 5: MAG chemical evidence. Left: SOS-evaluable topics as a proportion of the 1,000 requested topics. Right: compound-balanced mean SOS among eligible topics.

9 Conclusion

The accepted neural architecture scales successfully to 1,000 topics: it matches Tomotopy coherence, improves held-out likelihood, retains a broad topic inventory, and makes cached inference roughly 259 times faster. The remaining weaknesses are equally clear: Tomotopy still yields more diverse and active topics, broader chemical annotation, more high-confidence associations, and lower memory use. Claims are restricted to this completed seed-42 checkpoint; the next architecture work should target inventory and chemical coverage without sacrificing the newly closed coherence gap.

References

[1] J. J. J. van der Hooft et al. “Topic modeling for untargeted substructure exploration in metabolomics.” *Proceedings of the National Academy of Sciences*, 2016. https://pmc.ncbi.nlm.nih.gov/articles/PMC5137707/

[2] M. Torres Ortega et al. “Large-scale discovery and annotation of substructure patterns in mass spectrometry profiles.” *Nature Communications*, 2026. https://www.nature.com/articles/s41467-026-75038-0

[3] X. Wu et al. “Effective Neural Topic Modeling with Embedding Clustering Regularization.” *International Conference on Machine Learning*, 2023. https://proceedings.mlr.press/v202/wu23c.html

[4] A. Hoyle et al. “Are Neural Topic Models Broken?” *Findings of EMNLP*, 2022. https://aclanthology.org/2022.findings-emnlp.390/